Sprint Report

Unit code: EAT40006
Unit name Engineering Technology Project B (ENG/CS)-H1
Submission date: 21 Sep by 23:59
Report for: Sprint 5

1. Acknowledgment of Country

We acknowledge the Traditional Owners and Custodians of the lands on which this work was completed.

Rumeth Gunasingha and Thinidu completed this project while living and studying on the lands of the Wurundjeri Woi-wurrung People of the Kulin Nation. We pay our respects to Elders past and present and acknowledge their continuing connection to land, waters, culture, and community.

Contribution Summary (this sprint)

All team members should complete the following table together. You can provide a yes/no response or a brief comment where appropriate.

	Contribution	Initiatives	Communication					Respect
Team Member	Finished tasks in time or on time with an acceptable quality	Self-assigned tasks and proactively found solutions to problems	Attended all team meetings	Attended all supervisor meetings	Attended all client meetings	Replied every time within an acceptable timeframe	Kept the team updated about the status	The student respected others and listened to different opinions
Rumeth	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Thinidu	**Yes**	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Both team members contributed to literature review activities, simulation development, controller implementation, experimentation, documentation, and project planning. Communication was maintained through regular meetings, GitHub updates, and supervisor consultations.

2. Sprint plan

Sprint Goal

The primary objective of Sprint 5 was to transition from a theoretical project proposal into a functioning humanoid robot balancing framework. During Sprint 1, the team focused on project planning, literature review, and system design. During Sprint 5, the goal was to implement a simplified whole-body control framework capable of maintaining balance against external disturbances within a MuJoCo simulation environment.

Specifically, the sprint focused on:

- Building the humanoid simulation environment.
- Implementing centre-of-mass (CoM) tracking.
- Implementing a support-box safety filter.
- Developing inverse kinematics and posture control.
- Testing disturbance rejection through external pushes.
- Evaluating controller performance.

Scrum Roles

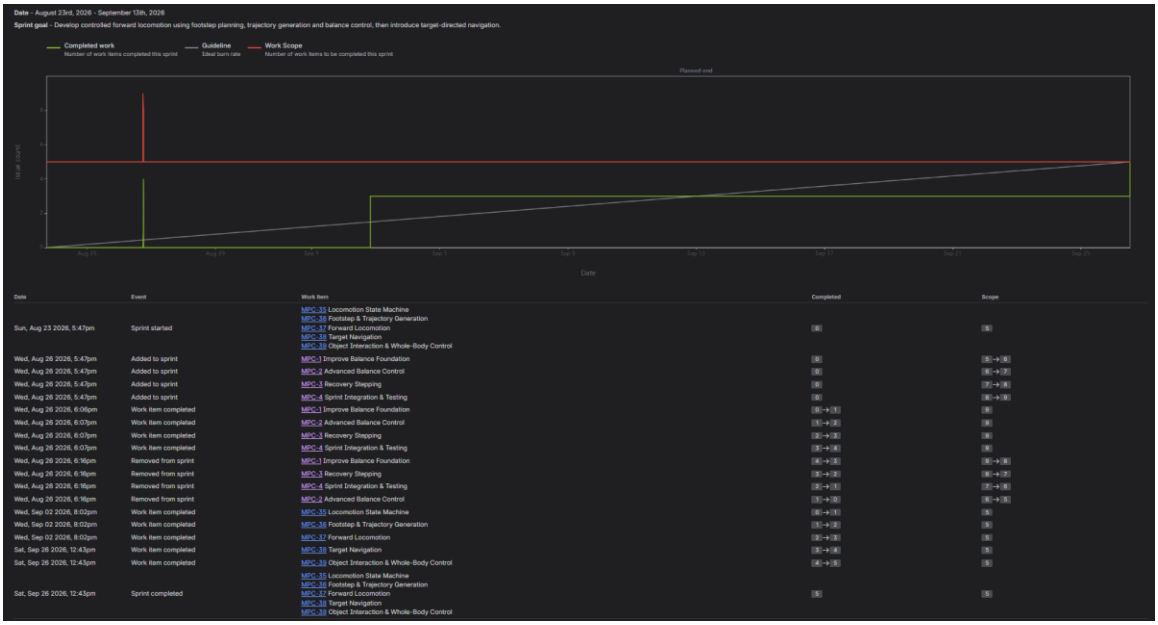
Scrum Master

- Rumeth Gunasingha
    - Responsibilities:
    - Sprint planning
    - Task allocation
    - Progress monitoring
    - Supervisor communication
    - Sprint review preparation

Development Team

- Rumeth Gunasingha
    - MuJoCo simulation
    - Whole-body controller
    - Data analysis
    - Report writing
- Thinidu
    - Literature review
    - Testing
    - Validation
    - Documentation
    - Results analysis

Sprint Backlog



Date	Event	Work item	Completed	Scope
Sun, Aug 23 2026, 5:47pm	Sprint started	MPC-35 Locomotion State Machine MPC-36 Footstep & Trajectory Generation MPC-37 Forward Locomotion MPC-38 Target Navigation MPC-39 Object Interaction & Whole-Body Control	0	5
Wed, Aug 26 2026, 5:47pm	Added to sprint	MPC-1 Improve Balance Foundation	0	5 → 6
Wed, Aug 26 2026, 5:47pm	Added to sprint	MPC-2 Advanced Balance Control	0	6 → 7
Wed, Aug 26 2026, 5:47pm	Added to sprint	MPC-3 Recovery Stepping	0	7 → 8
Wed, Aug 26 2026, 5:47pm	Added to sprint	MPC-4 Sprint Integration & Testing	0	8 → 9
Wed, Aug 26 2026, 6:06pm	Work item completed	MPC-1 Improve Balance Foundation	0 → 1	9
Wed, Aug 26 2026, 6:07pm	Work item completed	MPC-2 Advanced Balance Control	1 → 2	9
Wed, Aug 26 2026, 6:07pm	Work item completed	MPC-3 Recovery Stepping	2 → 3	9
Wed, Aug 26 2026, 6:07pm	Work item completed	MPC-4 Sprint Integration & Testing	3 → 4	9
Wed, Aug 26 2026, 6:16pm	Removed from sprint	MPC-1 Improve Balance Foundation	4 → 5	9 → 8
Wed, Aug 26 2026, 6:16pm	Removed from sprint	MPC-3 Recovery Stepping	3 → 2	8 → 7
Wed, Aug 26 2026, 6:16pm	Removed from sprint	MPC-4 Sprint Integration & Testing	2 → 1	7 → 6
Wed, Aug 26 2026, 6:16pm	Removed from sprint	MPC-2 Advanced Balance Control	1 → 0	6 → 5
Wed, Sep 02 2026, 8:02pm	Work item completed	MPC-35 Locomotion State Machine	0 → 1	5
Wed, Sep 02 2026, 8:02pm	Work item completed	MPC-36 Footstep & Trajectory Generation	1 → 2	5
Wed, Sep 02 2026, 8:02pm	Work item completed	MPC-37 Forward Locomotion	2 → 3	5
Sat, Sep 26 2026, 12:43pm	Work item completed	MPC-38 Target Navigation	3 → 4	5
Sat, Sep 26 2026, 12:43pm	Work item completed	MPC-39 Object Interaction & Whole-Body Control	4 → 5	5
Sat, Sep 26 2026, 12:43pm	Sprint completed	MPC-35 Locomotion State Machine MPC-36 Footstep & Trajectory Generation MPC-37 Forward Locomotion MPC-38 Target Navigation MPC-39 Object Interaction & Whole-Body Control	5	5

Artifacts Developed

The following artifacts were produced during Sprint 5:

- **Software Artifacts**
  - MuJoCo humanoid simulation environment
  - Centre-of-Mass tracking module
  - Support-box safety filter
  - Inverse kinematics module
  - Joint-space PD controller
  - Disturbance injection framework
  - Performance analysis scripts
- **Documentation**
  - Research report
  - Sprint report
  - GitHub repository updates
  - Experimental evaluation report

3. Quality management plan and outcome

Quality Objectives

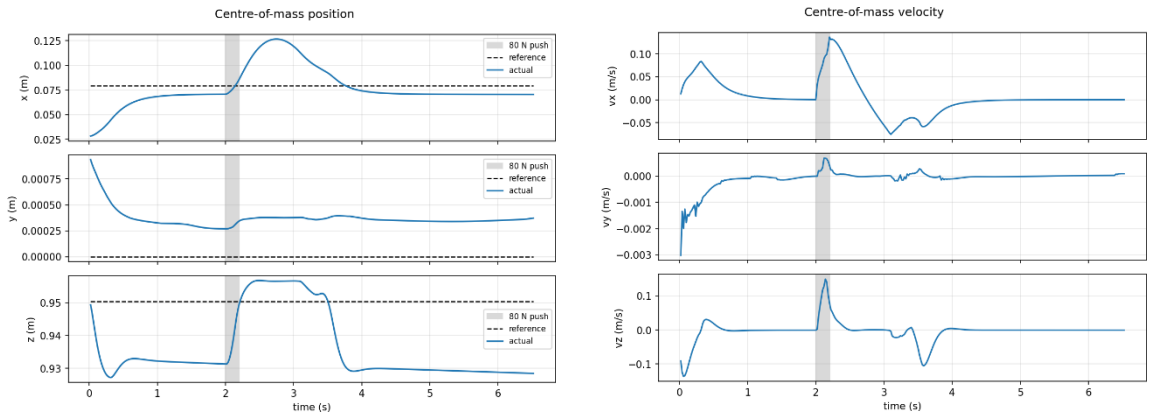
The project quality objectives were:

1. Maintain humanoid balance during normal standing.
2. Keep CoM inside support region.

3. Track reference posture accurately.
4. Respond safely to external disturbances.
5. Ensure controller stability.

Quality Metrics

Metric 1: CoM Position Error



- Target: Less than 5 cm deviation
- Measured: Maximum CoM tracking error: ≈ 4.7 cm
- The requirement was satisfied.

Metric 2: Standing Stability

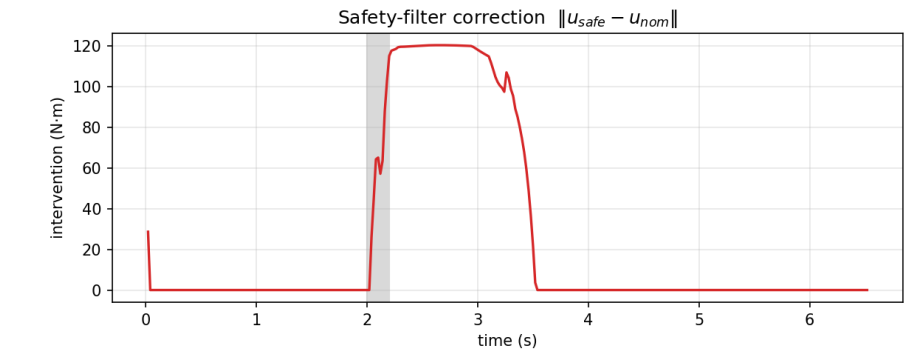
```
(.venv) PS C:\Users\rumet\Desktop\MPC new\project> python main_demo.py
[2026-09-26 20:01:21,190] INFO    main_demo | viewer open — close the window to stop. Push of 80 N at t = 2.0 s.
[2026-09-26 20:01:32,673] INFO    main_demo | t =  1.00 s | COM [0.068 0.000 0.932] m | |v| = 0.008 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:34,739] INFO    main_demo | t =  2.00 s | COM [0.071 0.000 0.931] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:34,740] INFO    main_demo | t = 2.00 s: applied [80.  0.  0.] N for 0.20 s on torso_link
[2026-09-26 20:01:37,987] INFO    main_demo | t =  3.02 s | COM [0.118 0.000 0.957] m | |v| = 0.060 m/s | intervention 117.30 Nm (relaxed)
[2026-09-26 20:01:40,253] INFO    main_demo | t =  4.02 s | COM [0.074 0.000 0.930] m | |v| = 0.012 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:42,233] INFO    main_demo | t =  5.02 s | COM [0.071 0.000 0.930] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:44,179] INFO    main_demo | t =  6.02 s | COM [0.071 0.000 0.929] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:45,993] INFO    main_demo | t =  7.02 s | COM [0.070 0.000 0.928] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:47,750] INFO    main_demo | t =  8.02 s | COM [0.070 0.000 0.927] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:49,615] INFO    main_demo | t =  9.02 s | COM [0.070 0.000 0.927] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 20:01:51,435] INFO    main_demo | t = 10.00 s | COM [0.070 0.000 0.926] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
```

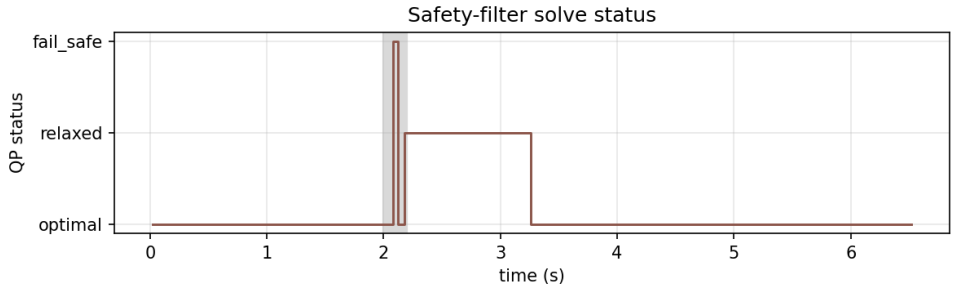
- Target: Robot remains upright without external disturbances.
- Result: The robot maintained stable standing posture before the push event.
- Peak pitch before push: 0.17°
- This indicates excellent stability.

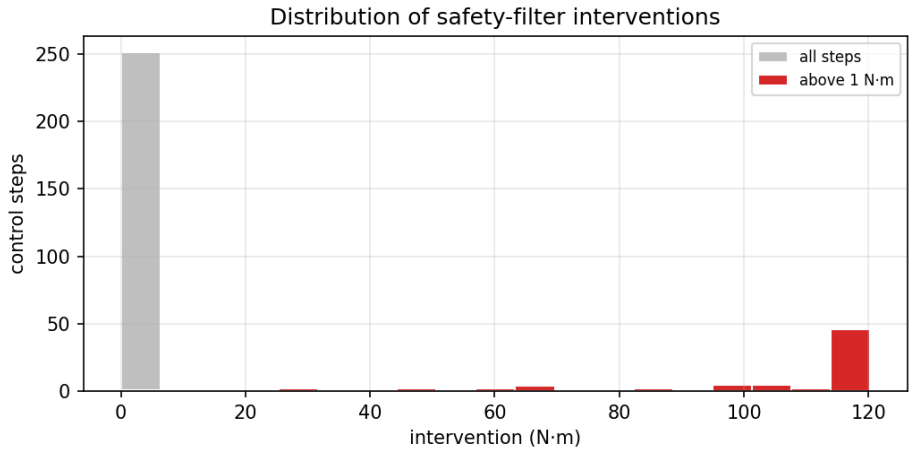
Metric 3: Disturbance Recovery

- Target:Recover from moderate pushes.
- Result:Robot survived an 80 N push.
- Push duration:0.2 seconds
- The robot recovered without falling.
-

Metric 4: Safety Filter Behaviour







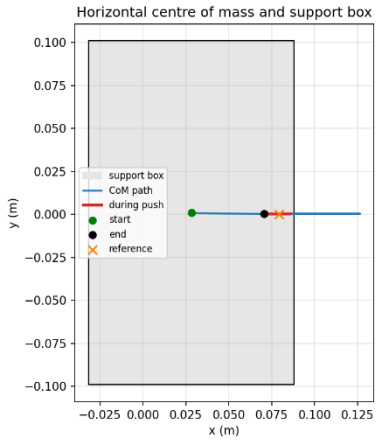
- Target:Only intervene when support constraints are violated.
- Result:The safety filter remained inactive during normal standing and became active only after the disturbance.
- Maximum intervention:120.27 N·m
- This demonstrates correct activation behaviour.

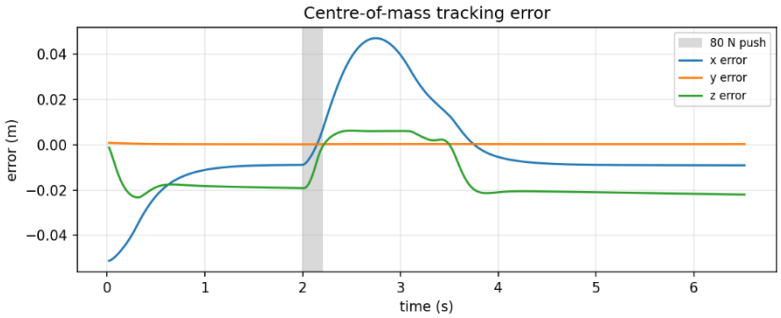
Testing Activities Completed

Unit Testing

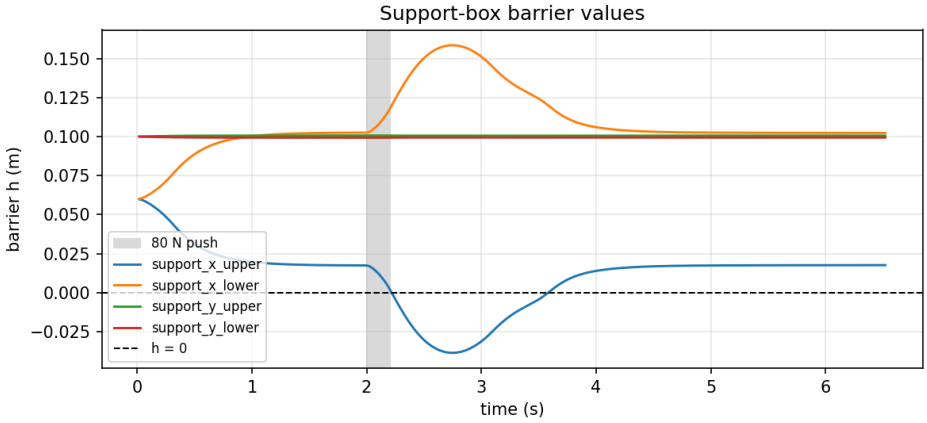
The following modules were tested independently:

- CoM computation

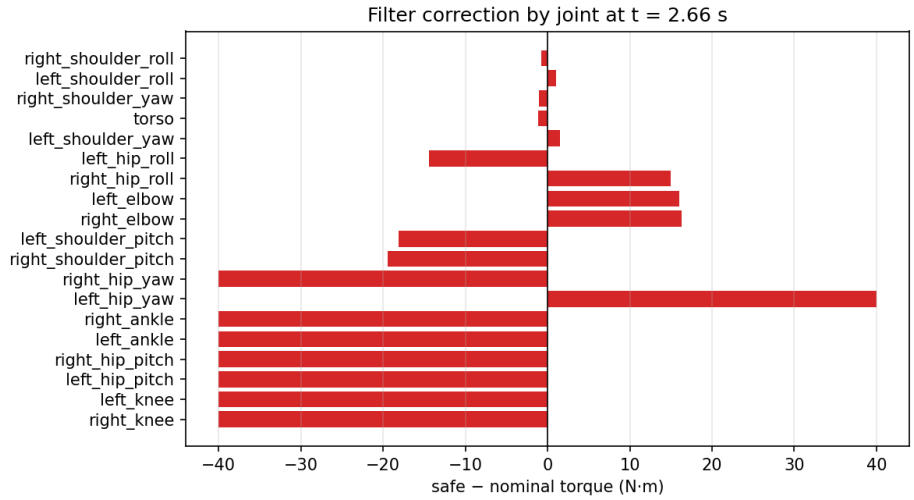




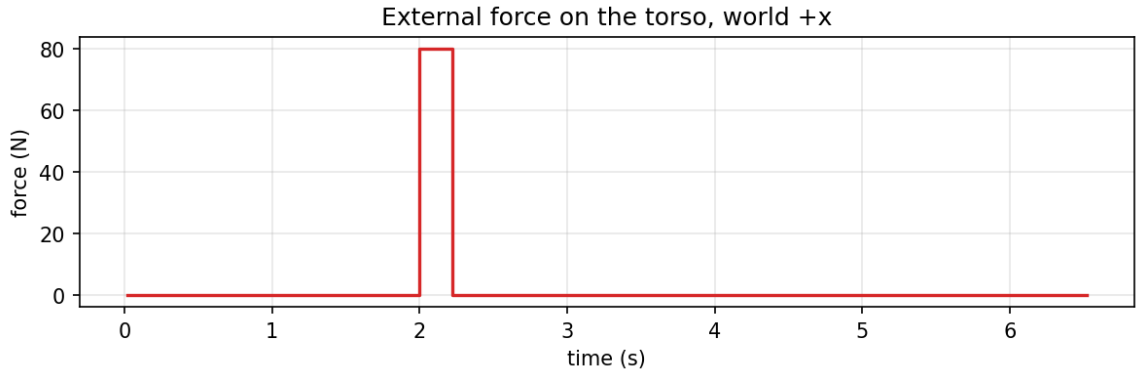
- Barrier function calculation



- Safety filter optimisation & Joint controller



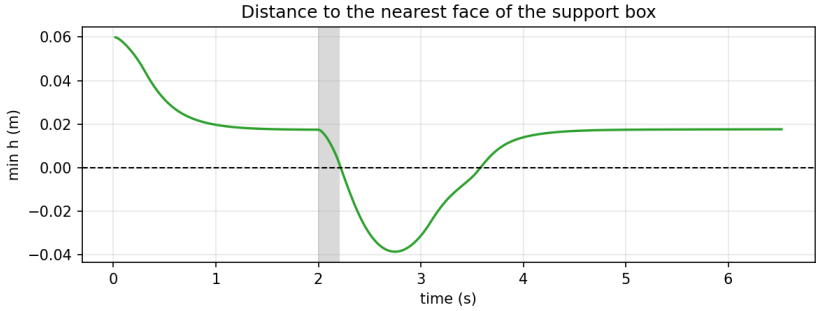
- Disturbance generator



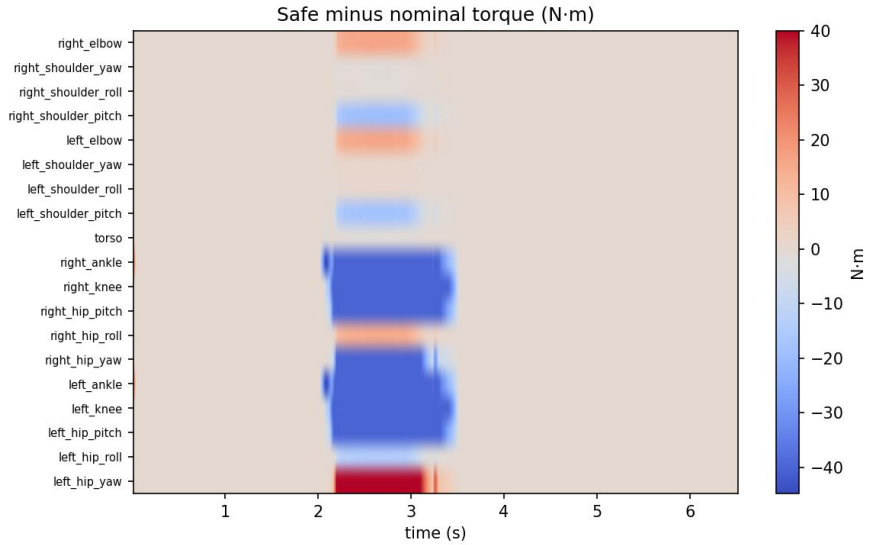
Integration Testing

Testing was performed on:

- IK + PD Controller
- Safety Filter + IK



- CoM Controller + Safety Filter



- Full Control Pipeline

Simulation Testing

Scenarios included:

- Standing without disturbance

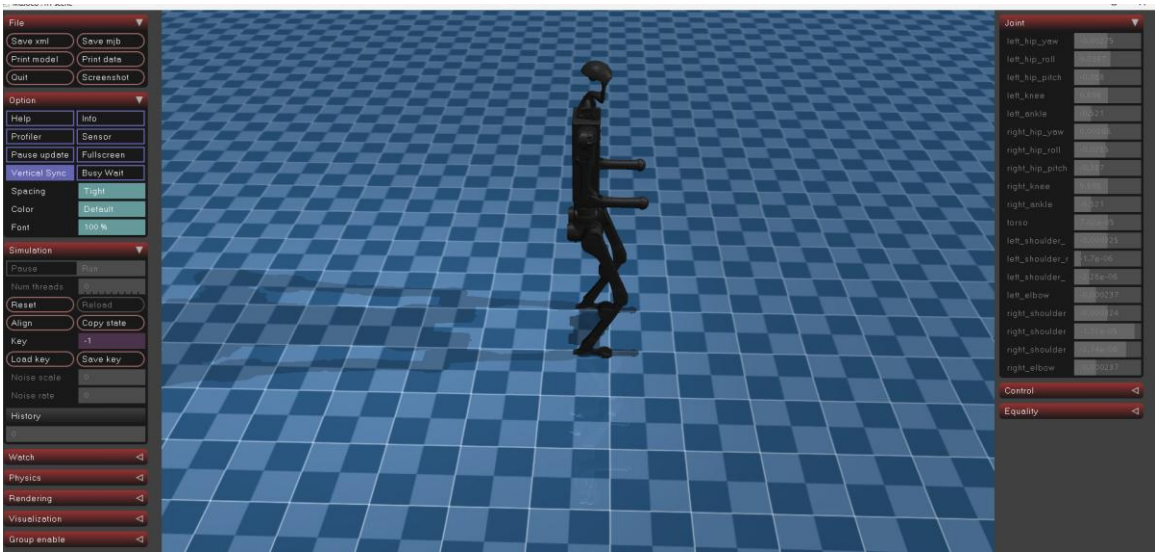
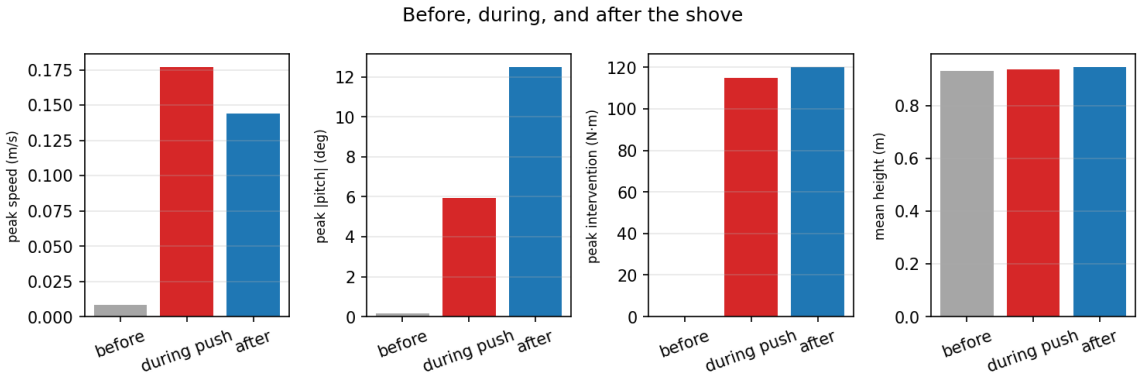


Figure 1 : Before Push



Figure 2: being pushed

- 80 N push & Controller recovery



- Long duration standing

Future Quality Activities

Future testing will include:

- Larger disturbance magnitudes
- Walking motions
- Uneven terrain
- Multi-contact balancing
- Manipulation tasks

4. Sprint Progress

Simulation Development

The MuJoCo environment was successfully configured and integrated with Python control code. The robot model includes a floating base and multiple actuated joints representing the lower and upper body.

The simulation operates at physics-step frequency while the control algorithm runs synchronously with the physics engine. This addressed stability issues encountered during early development where the controller updated too slowly.

Controller Development

- Three major control layers were developed:

Layer 1: Centre-of-Mass Control

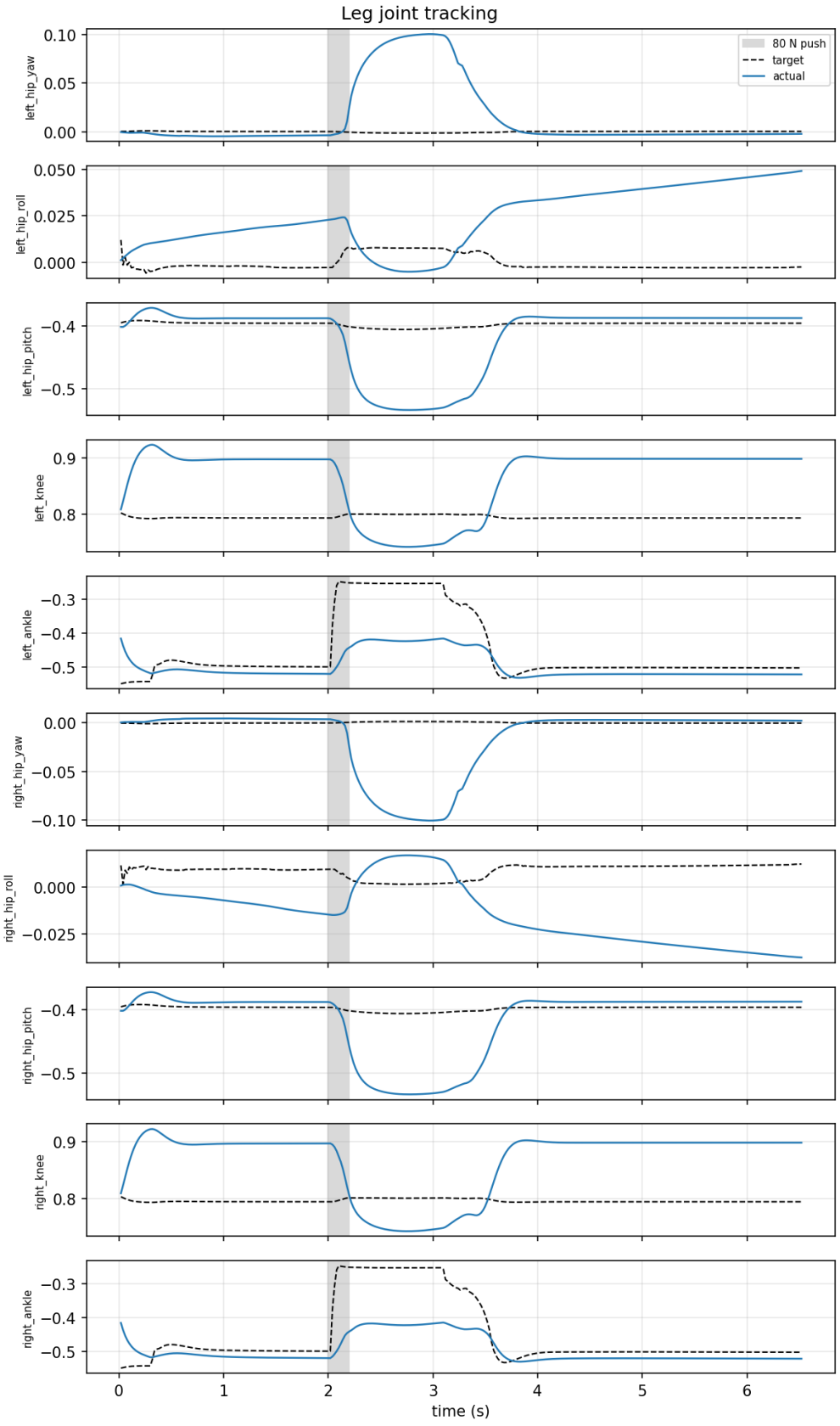
- The centre of mass acts as the primary balancing variable.
- The controller continuously monitors: (relevant images are attached in Matrics section
    - CoM position
    - CoM velocity
    - Support-box boundaries
- When the CoM approaches a boundary, corrective actions are generated.

Layer 2: Inverse Kinematics

- The inverse kinematics solver converts desired CoM motions into joint targets.
- During development, it was discovered that torso-pitch commands were being amplified through the leg kinematic chain. This caused excessive knee motion and immediate collapse.
- The solution was to remove unsupported torso rows from the Jacobian and only generate feasible lower-body motions.

Layer 3: Joint Controller

- A joint-level PD controller tracks desired positions.
- Initially the controller updated every 20 ms, causing the robot to behave like a compliant spring and collapse under load.
- The controller was modified to run every simulation step, significantly improving stability.

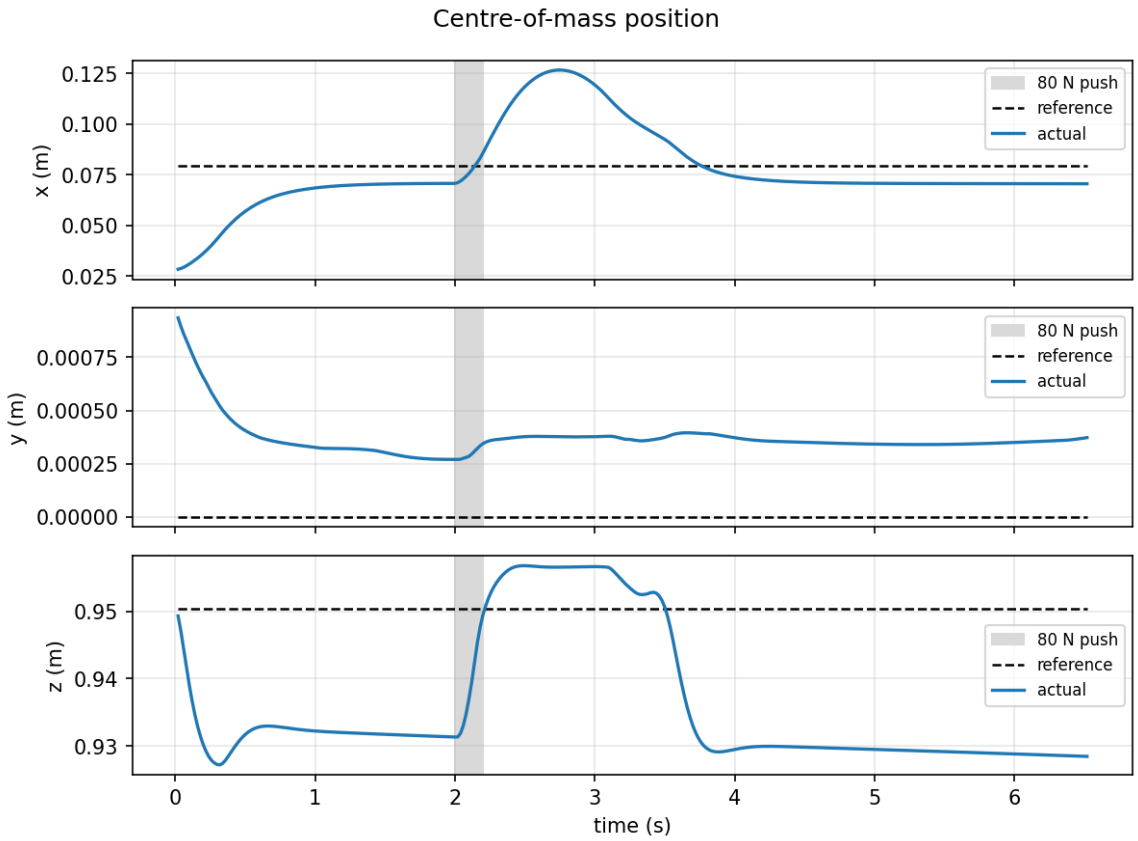

Leg joint tracking
80 N push
target
actual
left_hip_yaw
left_hip_roll
left_hip_pitch
left_knee
left_ankle
right_hip_yaw
right_hip_roll
right_hip_pitch
right_knee
right_ankle
time (s)

Experimental Results

An 80 N horizontal push was applied at 2 seconds for 0.2 seconds. The disturbance caused:

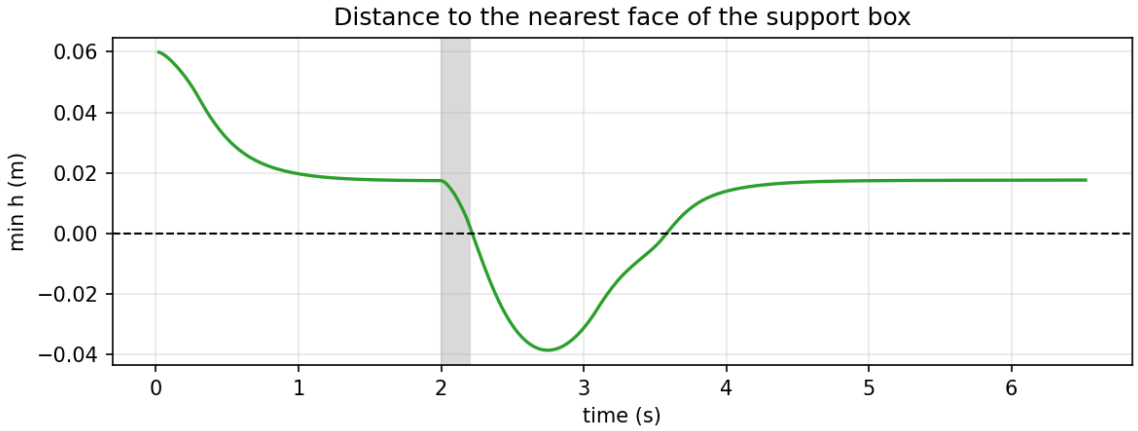
- Peak CoM speed: 0.177 m/s
- Peak pitch angle: 12.5°
- Peak forward CoM displacement: 0.127 m
- Peak safety intervention: 120 N·m

Centre-of-Mass Behaviour



- Before the push, the CoM remained close to the desired reference.
- During the disturbance, the CoM accelerated forward and temporarily moved outside the support-box boundary.
- The maximum forward position reached approximately 0.127 m while the reference position was 0.079 m. This corresponds to approximately 4.7 cm of forward tracking error.
- After the disturbance, the robot gradually returned toward the reference position.

Safety Filter Behaviour



- The support-box barrier function became negative during the push, indicating that the CoM moved outside the desired safe region.
- Minimum barrier value:
- −0.038 m
- When this occurred, the optimisation-based safety filter generated corrective torques.
- The intervention reached approximately 120 N·m and remained active for 76 control steps.
- This demonstrates that the safety layer correctly detected constraint violations and attempted recovery.

Robot Attitude

- The robot's pitch increased significantly following the push.
- Peak pitch: 12.5°
- The pitch response confirms that the robot experienced substantial forward rotation due to the external force. However, unlike earlier controller versions, the robot remained upright and eventually returned close to its nominal orientation.

Walking mode development

To extend the standing balance controller toward locomotion, a preliminary walking mode was implemented. Rather than introducing footstep generation or swing-leg motion, the initial objective was to verify that the robot could safely shift its weight forward while maintaining stability. This was achieved by moving the centre-of-mass (CoM) reference forward at a constant velocity of 0.05 m/s while both feet remained firmly planted on the ground.

The moving CoM reference was integrated into the existing MPC–PD–CBF control architecture. The reference trajectory begins at the robot's current centre of mass and progresses in the positive world-frame x-direction. To ensure stability, the reference is constrained to remain within the support region of the feet and stops approximately 2 cm before the toe boundary.

This prevents the robot from attempting to shift its weight beyond the available support polygon, which would otherwise increase the risk of tipping.

During implementation, it was observed that using the midpoint of the foot as the initial reference caused the robot to lean backward. The centre of mass initially rested closer to the heel region, while the rear boundary of the safety support box was positioned near the heel edge. Consequently, commanding the CoM toward the foot centre introduced an undesirable backward correction. This issue was resolved by initializing the moving reference directly at the measured CoM position and gradually advancing it forward.

Time	COM x	Reference x	Error x	Forward speed	Height
0.5 s	0.033 m	0.053 m	−0.020 m	0.037 m/s	0.928 m
2.0 s	0.104 m	0.128 m	−0.024 m	0.050 m/s	0.932 m
4.0 s	0.150 m	0.159 m	−0.009 m	0.000 m/s	0.931 m

Figure 3Headless forward CoM-shifting simulation

The Model Predictive Controller receives both the desired CoM position and the corresponding reference velocity over the prediction horizon. While the target is moving, the controller tracks the desired forward velocity of 0.05 m/s. Once the reference reaches the predefined limit near the toes, the velocity command becomes zero and the controller settles the robot into a stable forward-leaning stance. To accommodate the weight shift, the support box used by the Control Barrier Function safety filter was expanded to cover the full sole area, ensuring that safe forward movement could occur without unnecessary intervention.

```
(.venv) PS C:\Users\rumet\Desktop\MPC new\project> python main_demo.py --no-walk
[2026-09-26 22:49:52,498] INFO    main_demo | viewer open — close the window to stop. Push of 80 N at t = 2.0 s.
[2026-09-26 22:49:54,764] INFO    main_demo | t =  1.00 s | COM [0.068 0.000 0.932] m | ref x 0.079 m | error [-0.011 0.000 -0.018] m | |v| = 0.008 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:49:56,245] INFO    main_demo | t =  2.00 s | COM [0.071 0.000 0.931] m | ref x 0.079 m | error [-0.009 0.000 -0.019] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:49:56,260] INFO    main_demo | t = 2.00 s: applied [80.  0.  0.] N for 0.20 s on torso_link
[2026-09-26 22:49:57,751] INFO    main_demo | t =  3.02 s | COM [0.118 0.000 0.957] m | ref x 0.079 m | error [0.039 0.000 0.006] m | |v| = 0.060 m/s | intervention 117.30 Nm (relaxed)
[2026-09-26 22:49:59,274] INFO    main_demo | t =  4.02 s | COM [0.074 0.000 0.930] m | ref x 0.079 m | error [-0.006 0.000 -0.021] m | |v| = 0.012 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:50:00,778] INFO    main_demo | t =  5.02 s | COM [0.071 0.000 0.930] m | ref x 0.079 m | error [-0.009 0.000 -0.021] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:50:02,309] INFO    main_demo | t =  6.02 s | COM [0.071 0.000 0.929] m | ref x 0.079 m | error [-0.009 0.000 -0.022] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:50:02,653] INFO    main_demo | closed after 6.18 s, 309 steps. peak intervention 120.27 Nm. log: C:\Users\rumet\Desktop\MPC new\project\data\demo_log.npz
(.venv) PS C:\Users\rumet\Desktop\MPC new\project> python main_demo.py
[2026-09-26 22:54:39,163] INFO    main_demo | viewer open — close the window to stop. Walking at 0.05 m/s, both feet planted. CoM reference x from 0.028 m to 0.159 m.
[2026-09-26 22:54:42,066] INFO    main_demo | t =  1.00 s | COM [0.054 0.001 0.930] m | ref x 0.077 m | error [-0.023 0.000 -0.020] m | |v| = 0.047 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:43,515] INFO    main_demo | t =  2.00 s | COM [0.103 0.001 0.932] m | ref x 0.127 m | error [-0.024 0.000 -0.019] m | |v| = 0.050 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:45,078] INFO    main_demo | t =  3.02 s | COM [0.146 0.001 0.932] m | ref x 0.159 m | error [-0.013 0.000 -0.018] m | |v| = 0.014 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:46,625] INFO    main_demo | t =  4.02 s | COM [0.150 0.001 0.931] m | ref x 0.159 m | error [-0.009 0.000 -0.020] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:48,189] INFO    main_demo | t =  5.02 s | COM [0.150 0.001 0.930] m | ref x 0.159 m | error [-0.009 0.000 -0.020] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:49,709] INFO    main_demo | t =  6.02 s | COM [0.150 0.001 0.930] m | ref x 0.159 m | error [-0.009 -0.000 -0.021] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:51,289] INFO    main_demo | t =  7.02 s | COM [0.150 0.001 0.929] m | ref x 0.159 m | error [-0.009 -0.000 -0.021] m | |v| = 0.001 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:57,052] INFO    main_demo | t =  8.02 s | COM [0.150 0.001 0.929] m | ref x 0.159 m | error [-0.010 -0.000 -0.022] m | |v| = 0.000 m/s | intervention 0.00 Nm (optimal)
[2026-09-26 22:54:57,658] INFO    main_demo | closed after 8.34 s, 417 steps. peak intervention 0.00 Nm. log: C:\Users\rumet\Desktop\MPC new\project\data\demo_log.npz
```

Figure 4: COM shifting forward while maintaining balance. (walking mode vs not walking mode)

Experimental evaluation demonstrated successful forward weight transfer while maintaining balance. The centre of mass tracked the moving reference with an average lag of approximately 2 cm during motion and converged near the target position once the movement was completed. Throughout the experiment, the robot maintained a nearly constant height of approximately 0.93 m and retained stable foot contact with the ground. No stepping motion occurred, as both feet remained fixed for the duration of the test. The resulting behaviour confirms that the

controller can perform controlled CoM translation while preserving balance, providing a foundation for future development of dynamic stepping and full humanoid walking.

5. Sprint review (Critical review of the product)

Demonstration Session

A demonstration session was conducted with the supervisor and project stakeholders.

The team presented:

- Simulation architecture
- Controller structure
- Safety filter design
- Disturbance recovery experiments
- Experimental metrics

The demonstration was conducted professionally using simulation recordings. Questions were answered using supporting evidence from simulation logs and performance metrics.

Client Feedback

To be determined

Critical Analysis

Sprint objectives were largely achieved.

Major achievements include:

- Successful implementation of a balancing controller.
- Successful integration of safety filtering.
- Quantitative validation using disturbance tests.

The largest challenge was controller instability. Early versions of the controller collapsed immediately due to incorrect gravity compensation and inappropriate inverse-kinematics mappings.

The debugging process revealed that stability issues often originated from interactions between multiple controller layers rather than a single faulty component.

The final implementation successfully resolved these issues and demonstrated robust standing behaviour.

6. Retrospect (Critical review of the process)

Strengths

Regular Communication:

we maintained regular meetings and supervisor consultations throughout the sprint. These discussions helped us communicate technical problems, review simulation results, receive feedback, and decide the next development steps.

Incremental Development:

The controller was developed in separate components before full integration. Testing individual modules such as joint control, inverse kinematics, balance control, and walking components made debugging easier and helped identify where instability originated.

Evidence-Based Decision Making:

Controller changes were based on simulation results rather than assumptions. Robot posture, joint responses, COM behaviour, and stability were observed after each major modification, allowing the team to make more informed adjustments.

Challenges

Control-System Complexity:

Humanoid control contains many interacting components. Changes to the hip, knee, ankle, COM, or torso controller can affect the overall stability of the robot. This made integration more difficult than testing individual components.

Lack of Real Hardware:

Testing was performed entirely in MuJoCo because physical H1 hardware was unavailable. Simulation allowed safe and repeatable experiments, but real-world factors such as sensor noise, actuator behaviour, friction, and delays are not fully represented.

Computational Tuning:

Considerable experimentation was required to tune controller gains, IK parameters, safety constraints, and other control parameters. Small parameter changes sometimes caused significant differences in robot stability.

Cybersecurity and Ethics

The current project does not collect personal information because development is simulation-based. However, future deployment on a physical humanoid robot would require consideration of privacy, workplace safety, human-robot interaction, and secure communication between sensors and controllers.

Cybersecurity and Ethics

- The project currently does not collect personal information.
- However, future deployment of humanoid robots in industrial environments requires consideration of:
- Australian Privacy Principles (APPs)
- Workplace safety regulations
- Human-robot interaction safety
- Secure communication between controllers and sensors
- The team reviewed these requirements and will continue incorporating them into future development

7. Lessons learned (Critical review of sprint 5 experience and future plan)

Lessons Learned

Lesson 1

1. Whole-body control is highly sensitive to modelling assumptions.
2. Incorrect treatment of gravity compensation caused immediate instability despite otherwise correct controller logic.

Lesson 2

1. Safety filters cannot compensate for poor nominal control.
2. The nominal controller must already provide reasonable behaviour before optimisation-based safety methods can be effective.

Lesson 3

1. Controller update frequency is critical.
2. Running the joint controller at every physics step dramatically improved stability.

Lesson 4

1. Simulation visualisation is essential.
2. The diagnostic plots were critical for identifying instability sources and validating corrections.

Future Sprint Plan

The next sprint will focus on expanding the current system from static balancing and disturbance recovery toward more realistic humanoid locomotion behaviours. At present, the Unitree H1 robot can maintain balance while standing, recover from moderate external disturbances, and track a moving centre-of-mass reference using the MPC–PD–CBF control

architecture. The next phase aims to build upon this foundation and introduce dynamic balance and walking capabilities.

1. Implement Linear Inverted Pendulum Model (LIPM)

The current controller uses centre-of-mass tracking and balance recovery strategies but does not explicitly model humanoid walking dynamics. A Linear Inverted Pendulum Model (LIPM) will be implemented to provide a simplified representation of humanoid balance, allowing the system to predict centre-of-mass motion and generate more stable locomotion behaviours.

2. Implement Capture-Point Control

The current system recovers from disturbances by regulating the centre of mass while both feet remain planted. Capture-point control will be introduced to estimate where the robot must place its foot in order to prevent a fall. This provides a theoretical framework for transitioning from static balancing to dynamic stepping recovery.

3. Add Stepping Recovery Strategies

At present, disturbance rejection relies entirely on ankle and body motion. If disturbances become too large, the robot eventually exceeds its support limits. Stepping recovery strategies will allow the robot to take corrective steps when balance cannot be maintained using fixed-foot control alone, improving robustness against larger external forces.

4. Improve Disturbance Rejection Capability

Current experiments demonstrate successful recovery from moderate pushes. Future work will investigate larger disturbance magnitudes, different push directions, and longer force durations. Controller parameters and safety constraints will be refined to improve recovery performance under more challenging conditions.

5. Extend Support Regions for Dynamic Balance

The current Control Barrier Function uses a simplified support region based on fixed foot placement. Future work will extend this concept to dynamic support regions that change according to the active support foot, enabling safer weight transfer and walking motions while maintaining formal safety guarantees.

6. Implement Walking Trajectory Generation

The robot currently performs forward centre-of-mass shifting while both feet remain on the ground. Walking trajectory generation will introduce swing-foot trajectories, footstep planning, and weight-transfer mechanisms to enable continuous forward locomotion. This will represent the first transition from static balance control to true humanoid walking.

7. Investigate Model Predictive Control (MPC) Enhancements

A simplified MPC framework is currently used for centre-of-mass tracking and posture regulation. Future work will investigate more advanced whole-body MPC formulations capable of simultaneously coordinating balance, locomotion, and disturbance rejection while respecting actuator and safety constraints.

8. Evaluate Safety Guarantees Under Larger Disturbances

The current safety filter successfully limits unsafe behaviour during moderate disturbances. Additional experiments will be conducted to evaluate whether the Control Barrier Function can maintain forward invariance of the safe set under stronger pushes and more dynamic motions, particularly during stepping and walking.

9. Compare Controller Performance Against Published Literature

Experimental results obtained from the developed system will be compared against published humanoid robotics research. Performance metrics such as centre-of-mass tracking error, disturbance recovery time, maximum body pitch angle, and stability margins will be used to assess the effectiveness of the proposed control architecture relative to existing approaches.